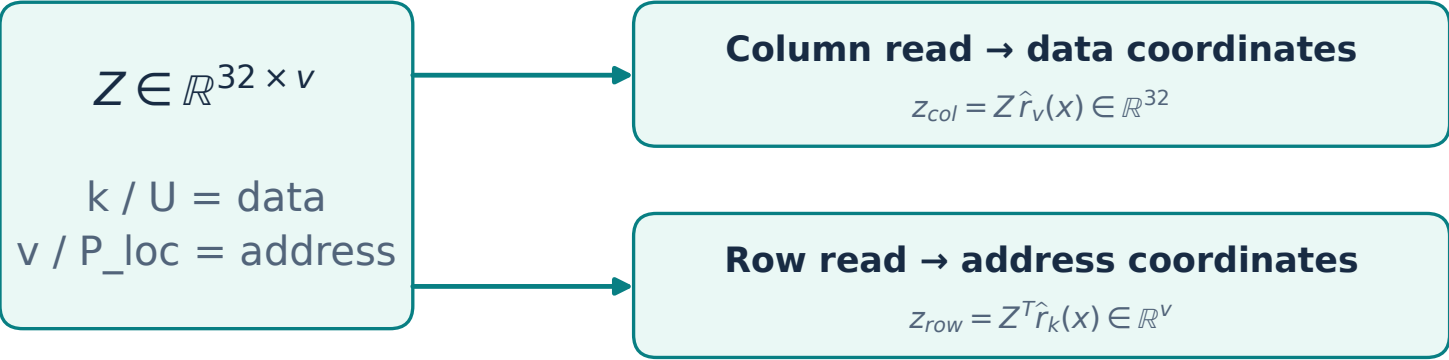


BAM read / write mechanics

Companion to the Transformer interaction diagram · same SharedRead LLF configuration

A / BILATERAL READ

One matrix, two directions



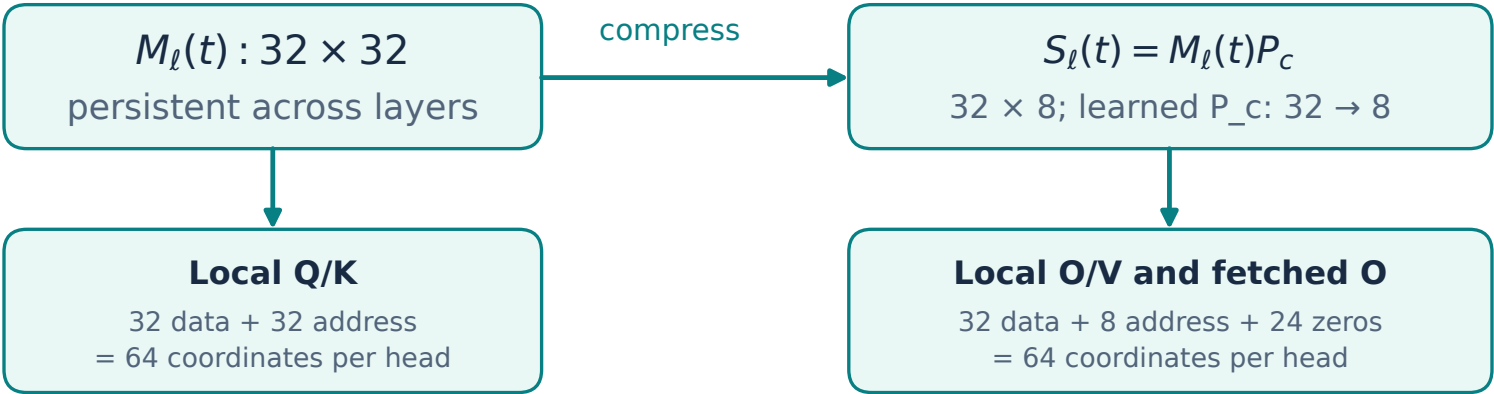
$\hat{r} = \text{RMS}(r)$ · row and column keys normalized separately

Read keys, routing weights and sigmoid gates are functions of x.

The persistent matrix stream has no whole-matrix RMS normalization.

B / EXACT WIDTHS

Full state and compressed read view



Compression changes only the read view; every write updates full M.

No learned 8 → 32 decoder: compressed row results occupy slots 32:40.

C / LOCAL QK & SHARED O/V

What is shared, and what is independent?

Local Q/K: one basis per read side, separately for Q and K.

$$\Delta Q_n = [a_{n,col}^Q z_{col}^Q ; a_{n,row}^Q z_{row}^Q]$$

Here z includes side gates; signed, RMS-normalized head weights route it.

K uses the same construction with its own keys, gates and head weights.

Local O/V: 16 read heads share one pre-gate result between destinations.

$$\Delta V_n = G_{V,n}(x) \odot R_n, \quad \Delta O_n = G_{O,n}(x) \odot R_n$$

$G_{V,n}, G_{O,n}$ each have row / column factors $2\sigma(\cdot)$. No cross-layer weight tying.

D / FETCH ROUTE & WRITE

Reuse attention for transport; write locally

$$w_t = \text{RMS}(W_{mix} x_t) / \sqrt{16}$$

$$\beta_{ts} = \sum_n w_{tn} A_n(t, s) \quad (s < t), \quad \beta_{tt} = 1$$

$$\bar{S}_t = S_\ell(t) + \sum_{s < t} \beta_{ts} S_\ell(s)$$

Signed route; causal / segment mask inherited from MHA; no second softmax.

$$P_{loc}(x) = W_{up} \text{GELU}(W_{down} x) + b$$

1024 → 256 → (16 × 32); one address factor per MHA head.

Write uses O before W_O; retention = 1; data and address factors use RMS.

E / READING THE TWO STREAMS

M carries depth history; MHA supplies token-to-token communication

L layer: M(t) → Q/K, V and O at token t; injected V is then transported by ordinary MHA.

F layer: the existing MHA map transports S(s) to t; a query-conditioned read converts the fetched matrix into O.

Both: fused O(t) supplies write data, while x(t) supplies the address and write gate; the FFN updates only the residual stream.